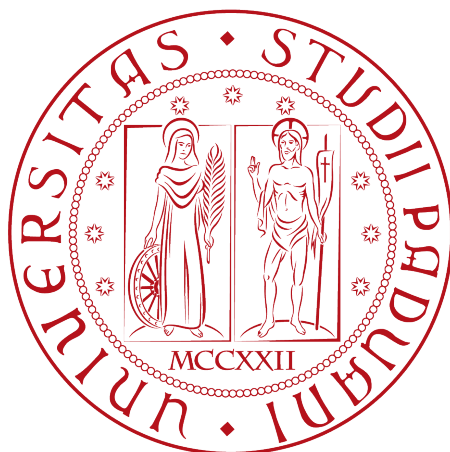


University of Padua

DEPARTMENT OF MATHEMATICS “TULLIO LEVI-CIVITA”

MASTER DEGREE IN COMPUTER SCIENCE



GeoMedia: A Geographic Framework for Location-Based Content Sharing

Master's Thesis

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Summary

Over the past few decades, the use of mobile devices has increased drastically, almost replacing desktop software and web browsers. This shift has been driven not only by the widespread adoption of smartphones, but also by the availability of integrated sensors such as GPS, gyroscopes, and Bluetooth, which allows a richer and more context-aware user experiences. By leveraging these technologies, Online Social Media platforms have been able to provide more advanced and sophisticated features; consequently collecting larger amounts of metadata, interaction and daily engagement.

This thesis presents “GeoMedia”, an open-source framework designed for sharing multimedia content — such as text, images, audio, and files — anchored to real-world geographic locations. By leveraging GPS data available on mobile devices, the application provides an interactive map through which users can explore and publish digital content, creating a seamless integration between the digital and physical worlds. Furthermore, the framework offers several features and cover various use cases, including restricting content visibility within customizable geographic areas, enabling time-sensitive content, applying user-based restrictions for content creation and visualization, and supporting sequential visibility between published posts. Furthermore, analyzes the motivations behind GeoMedia, its critical aspects and design considerations, describes its system architecture and implementation through a comparison of different technological paradigms. Finally, it explores potential applications and outlines possible future developments of the framework.

An previous version of this framework has been presented at “GoodIT ’25: International Conference on Information Technology for Social Good” (September 3 - 5, 2025 - Antwerp, Belgium) [1]

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Chapter 1

Introduction

1.1 Overview

Over the last two decades, mobile devices such as smartphones and tablets have become more and more present in everyday life, shaping people's habits and changing the way they interact with technology and digital applications, almost replacing traditional desktop software and web browsing.

There are many examples of this trend in modern society. For instance, when planning a trip or driving, smartphones assist users through apps like Google Maps or Waze. These apps don't just provide directions—they also improve the experience with real-time traffic updates, road event notifications, and information about available parking. In this way, the surrounding environment is enriched with digital details. Another example is Google Lens, which can identify objects from a photograph, or Shazam that can recognize a song through the device's microphone.

The technology integration has become an important key even in workplaces and public spaces, from the restaurant that provide a digital menu by a tablet, to customers checking a shopping lists. Similarly, many museums allowing visitors to scan QR codes to access audio-guided tours or unlock Augmented Reality functionalities that enhance the experience.

The wider adoption of smartphones has been driven by two main factors:

1. **Hardware:** the increase in computational performance, battery life, fast connectivity, and the integration of several built-in sensors such as gyroscopes, GPS, NFC, light sensors, cameras, and vibration motors. These enable applications to interact with different fields such as health, education, and, through motion-sensors or cameras, a coexistence with physical world, creating a more immersive and user experience.
2. **Software:** the development and presence of mobile applications has played a crucial role. In fact, platforms such as 'Windows Phone' were eventually abandoned due to the lack of applications compared to operating system like 'iOS' and 'Android'. This market highly influenced application design, pushing developers towards a 'mobile-first' approach and encouraging the exploitation of device sensors to create a more unique and sophisticated applications.

1.2 Social networks

Alongside the adoption of mobile technologies, social networks have emerged as a main role in the technology market, placing their position among the most used applications globally. These platforms have totally changed the way people communicate digitally, by providing real-time messaging capabilities in both one-to-one and one-to-many formats, while also

allowing users to share any kind of multimedia, including photos, videos, audio recordings, and other files. Applications such as WhatsApp, Telegram, Instagram, and Facebook have attracted billions of users worldwide, becoming deeply embedded in daily communication practices across diverse social, professional, and public contexts.

By facilitating the instantaneous exchange of information and supporting multiple forms of interaction, social networks have not only expanded the scope of personal and professional communication but have also transformed the ways in which knowledge, ideas, and content are disseminated and consumed. These platforms are predominantly user-centric, leveraging personalized recommendation algorithms that tailor content according to users' interests, interactions, and browsing behavior. As a result, they foster engagement while facilitating the creation and maintenance of digital relationships across diverse communities. By bridging geographical distances and enabling individuals to connect around shared interests, social networks have contributed to the development of increasingly interconnected digital ecosystems. Beyond their communication functions, these platforms have evolved into important channels for entertainment, news consumption, content creation, and professional networking, reflecting their multifaceted role in contemporary society. Consequently, social media has redefined the landscape of digital connectivity, creating new opportunities for collaboration, engagement, and participation on a global scale.

Chapter 2

Related works

SOCIAL MEDIA

- snapchat - instagram - google maps / trip advisor - foursquare - jagat - auglinn - wallame??

2.1 Instagram and Snapchat

2.2 Google Maps

2.3 Strava

2.4 WallaMe

Chapter 3

GeoMedia

3.1 Idea

3.2 Functionalities

Chapter 4

Conclusions

- 4.1 Consuntivo finale
- 4.2 Raggiungimento degli obiettivi
- 4.3 Conoscenze acquisite
- 4.4 Valutazione personale

“Life is really simple, but we insist on making it complicated”

— Confucius

Ringraziamenti

Innanzitutto, vorrei esprimere la mia gratitudine al Prof. Claudio Enrico Palazzi, relatore della mia tesi, per l'aiuto e il sostegno fornitomi durante la stesura del lavoro.

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Padua, May 2026

Alberto Morini

Appendix A

Appendice A

Citazione

Autore della citazione

Acronimi e abbreviazioni

API [Application Program Interface](#). [9](#)

UML [Unified Modeling Language](#). [9](#)

Glossario

API in informatica con il termine *Application Programming Interface API* (ing. interfaccia di programmazione di un'applicazione) si indica ogni insieme di procedure disponibili al programmatore, di solito raggruppate a formare un set di strumenti specifici per l'espletamento di un determinato compito all'interno di un certo programma. La finalità è ottenere un'astrazione, di solito tra l'hardware e il programmatore o tra software a basso e quello ad alto livello semplificando così il lavoro di programmazione. [8](#)

UML in ingegneria del software *UML, Unified Modeling Language* (ing. linguaggio di modellazione unificato) è un linguaggio di modellazione e specifica basato sul paradigma object-oriented. L'*UML* svolge un'importantissima funzione di “lingua franca” nella comunità della progettazione e programmazione a oggetti. Gran parte della letteratura di settore usa tale linguaggio per descrivere soluzioni analitiche e progettuali in modo sintetico e comprensibile a un vasto pubblico. [8](#)

Bibliografia

Riferimenti bibliografici

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- [1] *GoodIT 2025 - International Conference on Information Technology for Social Good*. 2025. URL: <https://goodit2025.org> (cit. on p. [ii](#)).
- [3] *Manifesto Agile*. URL: <http://agilemanifesto.org/iso/it/>.